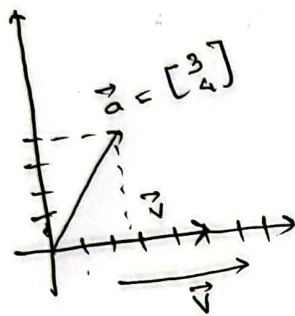
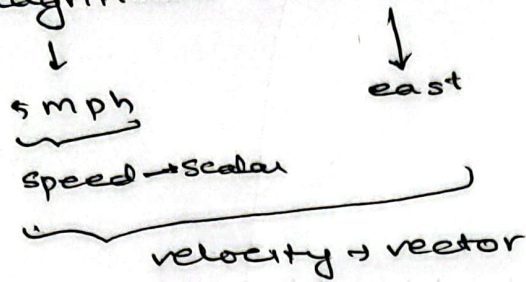


Unit-1: Vector and spaces

Lesson 1: vectors

vector intro for linear algebra

vector - magnitude + direction



$$\vec{v} = (5, 0) = \begin{bmatrix} 5 \\ 0 \end{bmatrix}$$

Real coordinate spaces

$\mathbb{R}^2$ } real coordinate space
 $\mathbb{R}^2$ } d-dimensional

$\downarrow$
all possible real-valued 2-tuple

$\mathbb{R}^3$ - 3D real coordinate space

$\downarrow$
real valued 3-tuples

$$\vec{x} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} -1 \\ 5 \\ 3 \end{bmatrix}$$

$$\vec{x} \in \mathbb{R}^3$$

$$\vec{b} \in \mathbb{R}^3$$

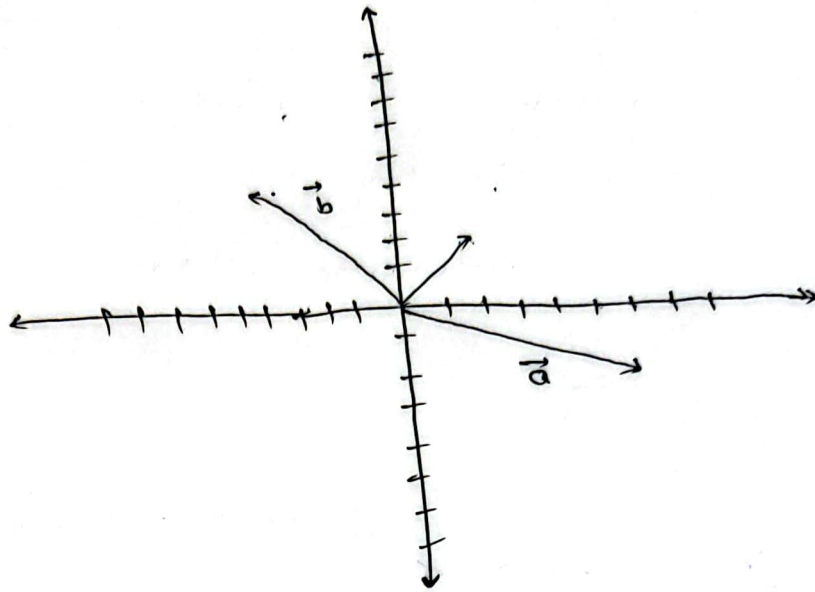
$\mathbb{R}^n$ - n-dimensional real coordinate space

Adding vectors algebraically & graphically

$$\vec{a} = \begin{bmatrix} 1 \\ -2 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} -4 \\ 4 \end{bmatrix}$$

$$\vec{a}, \vec{b} \in \mathbb{R}^2$$

$$\vec{a} + \vec{b} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$



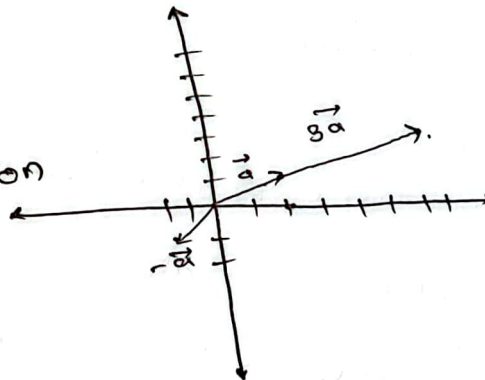
Multiplying a vector by a scalar

$$\vec{a} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \quad \text{scalar}$$

$$3\vec{a} = 3 \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 6 \\ 3 \end{bmatrix} \quad \text{same direction}$$

$$-1\vec{a} = -1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} \\ = \begin{bmatrix} -2 \\ -1 \end{bmatrix}$$



vector examples

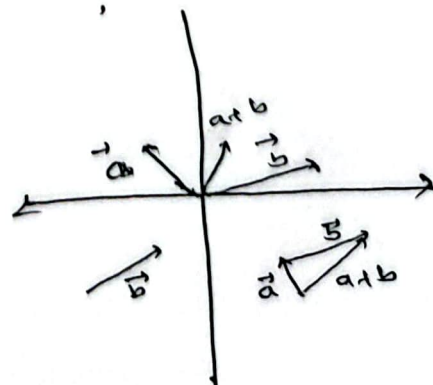
$\mathbb{R}^2$ ordered 2-tuples

(x_1, x_2)

$x_1 \in \mathbb{R}, x_2 \in \mathbb{R}$

$$\vec{a} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} \quad \vec{b} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

$$\vec{a} + \vec{b} = \begin{bmatrix} -1+3 \\ 2+1 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$



any point in $\mathbb{R}^2$

$\rightarrow (x_1, x_2)$

$(x_1 - 1, x_1 + 2)$

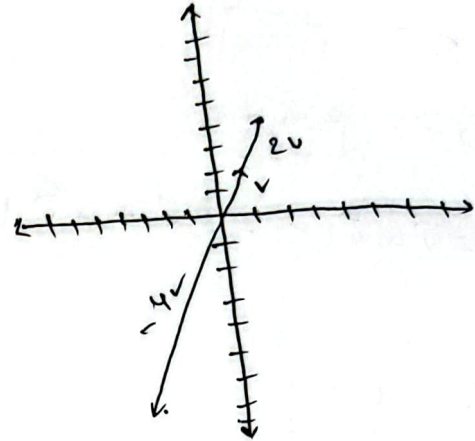
$(-4, 4)$

$(-4 - 1, 4 + 2)$

$(-5, 6)$

$$v = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \quad -4v = \begin{bmatrix} -4 \\ -8 \end{bmatrix}$$

$$2v = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

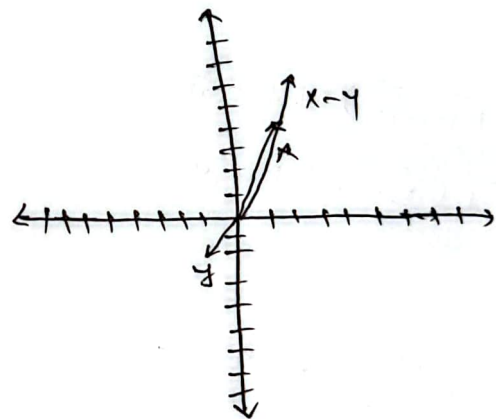


$$x = \begin{bmatrix} 2 \\ 4 \end{bmatrix}, y = \begin{bmatrix} -1 \\ -2 \end{bmatrix}$$

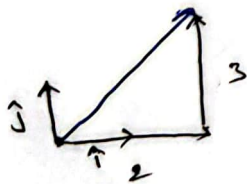
$$x - y = x + (-1)y$$

$$= \begin{bmatrix} 2 \\ 4 \end{bmatrix} + \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 3 \\ 6 \end{bmatrix}$$



unit vectors Intro



$$\vec{v} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

$$\vec{v} = (2, 3)$$

unit vectors

$$\hat{i} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\hat{j} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\vec{v} = 2\hat{i} + 3\hat{j}$$

$$\vec{b} = -1\hat{i} + 4\hat{j} = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

$$\begin{aligned} \vec{v} + \vec{b} &= (2 + (-1))\hat{i} + (3 + 4)\hat{j} \\ &= \hat{i} + 7\hat{j} \\ &= \begin{bmatrix} 1 \\ 7 \end{bmatrix} \end{aligned}$$

parametric representation of lines

$$\vec{v} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \leftarrow \text{position vectors}$$

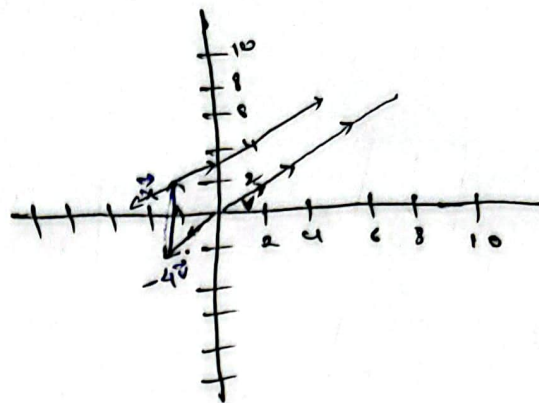
$$S = \{c\vec{v} \mid c \in \mathbb{R}\} \leftarrow \text{set of colinear vectors}$$

$$2\vec{v} = \begin{bmatrix} 4 \\ 2 \end{bmatrix} \quad \vec{x} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

$$1.5\vec{v} = \begin{bmatrix} 3 \\ 1.5 \end{bmatrix}$$

$$-2\vec{v} = \begin{bmatrix} -4 \\ -2 \end{bmatrix}$$

$$-2\vec{v} + \vec{x} = \{ \vec{x} + t\vec{v} \mid t \in \mathbb{R} \}$$

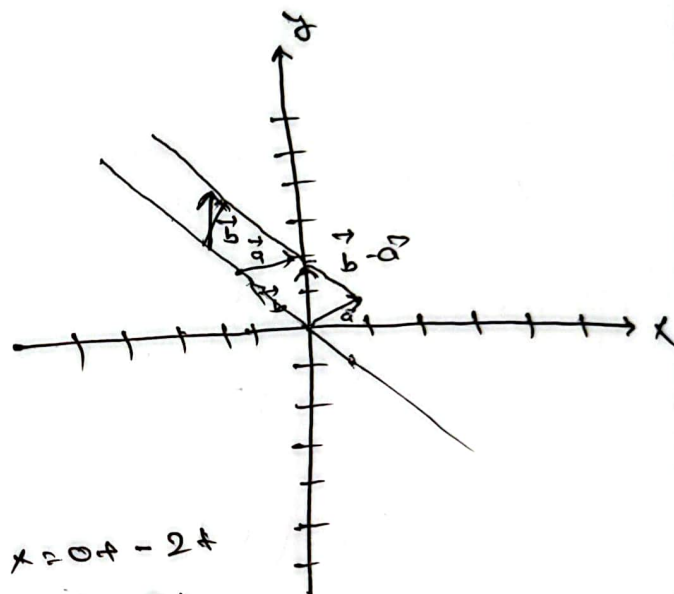


$$\vec{a} = \begin{bmatrix} 2 \\ 1 \end{bmatrix},$$

$$\vec{b} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\vec{b} + t(\vec{b} - \vec{a}) \mid t \in \mathbb{R}$$

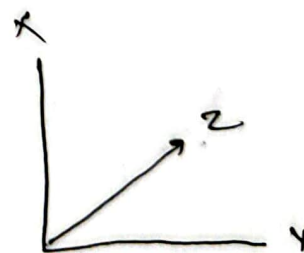
$$\vec{a} + t(\vec{b} - \vec{a}) \mid t \in \mathbb{R}$$



$$L = \left\{ \begin{bmatrix} 0 \\ 3 \end{bmatrix} + t \begin{bmatrix} -2 \\ 2 \end{bmatrix} \mid t \in \mathbb{R} \right\} \begin{matrix} \swarrow \text{x slope} \\ \searrow \text{y coordinate} \end{matrix} \quad \begin{matrix} x = 0 + -2t \\ y = 3 + 2t \end{matrix}$$

$$\vec{L}_1 = \begin{bmatrix} L_1 \\ L_2 \end{bmatrix} \begin{matrix} \swarrow \text{x coordinate} \\ \searrow \text{y coordinate} \end{matrix}$$

$$\vec{p}_1 = \begin{bmatrix} -1 \\ 2 \\ 7 \end{bmatrix} \quad \vec{p}_2 = \begin{bmatrix} 0 \\ 3 \\ 4 \end{bmatrix} \in \mathbb{R}^3$$



$$L = \{ \vec{p}_1 + t(\vec{p}_1 - \vec{p}_2) \mid t \in \mathbb{R} \}$$

$$\vec{p}_1 - \vec{p}_2 = \begin{bmatrix} -1 \\ -1 \\ 3 \end{bmatrix}$$

$$L = \left\{ \begin{bmatrix} -1 \\ 2 \\ 7 \end{bmatrix} + t \begin{bmatrix} -1 \\ -1 \\ 3 \end{bmatrix} \mid t \in \mathbb{R} \right\}$$

$$x = -1 + -1t, \quad y = 2 + -1t, \quad z = 7 + 3t$$